[파이썬 자료형과 연산자 응용, 입출력 함수의 활용]

- 숫자형과 기본 연산자의 활용
- 화면 입출력, 파일 입출력
- 개발 환경 익숙해 지기
    - Python 인터프리터 사용하기
    - VisualStudio Code 사용하기

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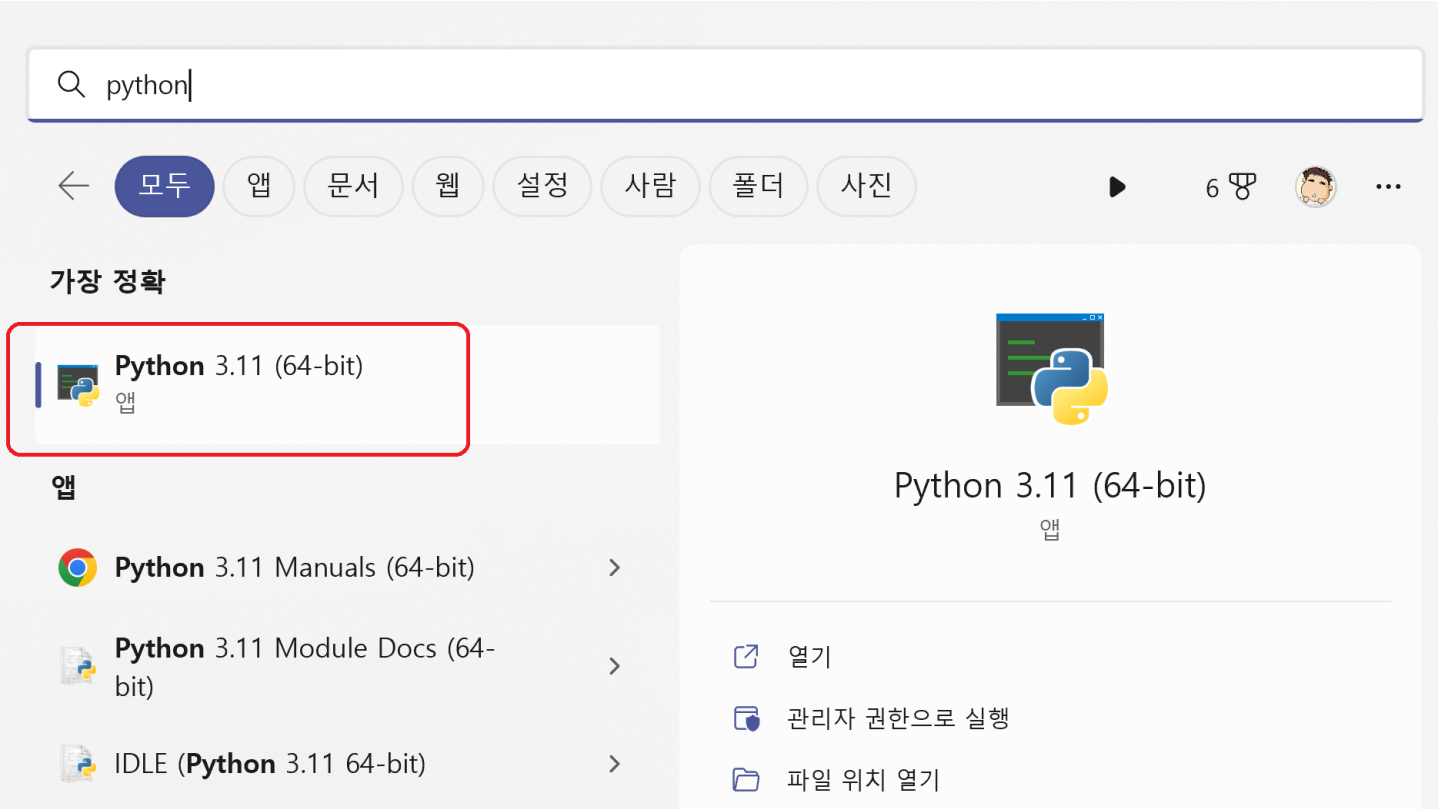
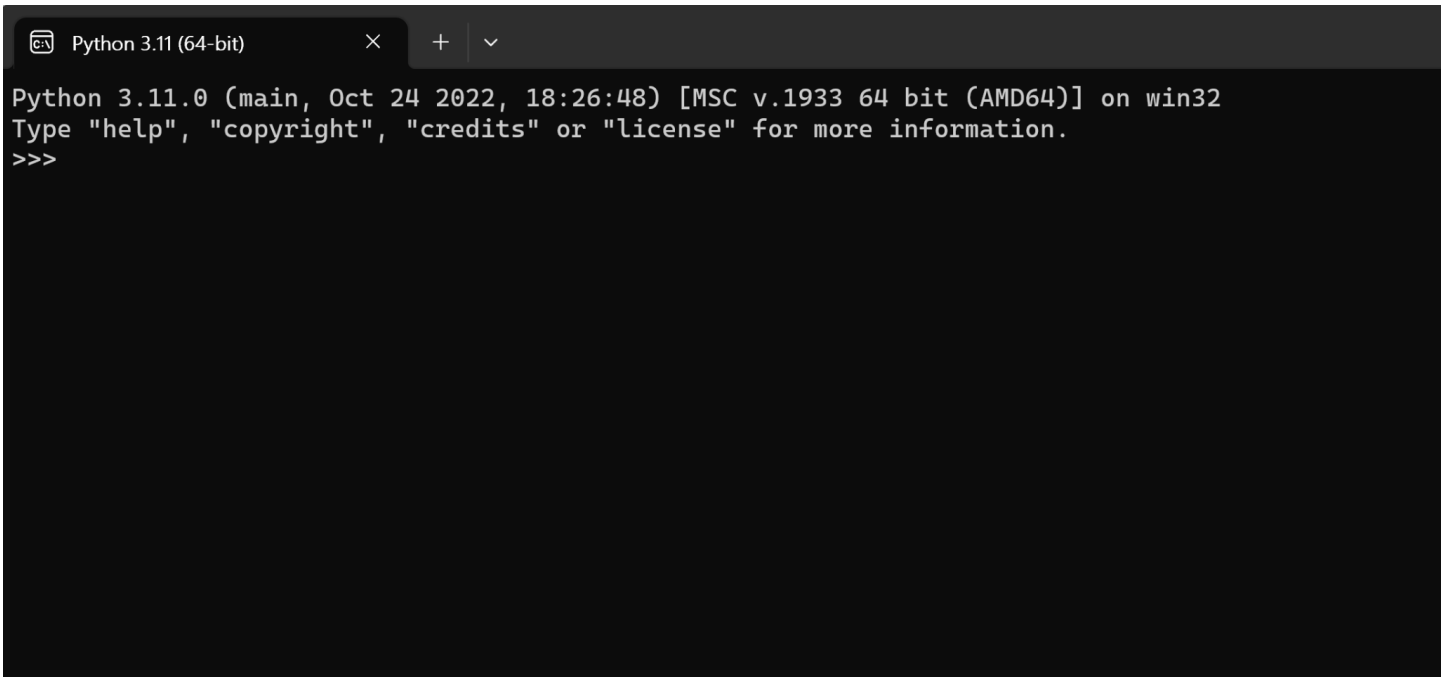
1. 파이썬 자료형과 연산자 응용

<*Python* 인터프리터를 이용한 자료형과 연산자 응용>

- Python 인터프리터 실행하기





- 자료형과 연산자 코딩의 예

● 정수, 실수, 16진수의 사용

```
Python 3.11 (64-bit)
Python 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> variable = 123
>>> print(variable)
123
>>> variable = 1.23E2
>>> print(variable)
123.0
>>> variable = 1.23e3
>>> print(variable)
1230.0
>>> variable = 1.23e-2
>>> print(variable)
0.0123
>>> variable = 0x1f
>>> print(variable)
31
>>> variable = 0x1F
>>> print(variable)
31
>>>
```

● 숫자형 데이터를 위한 연산자의 사용

```
Python 3.11 (64-bit)
Python 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> variable = 3 / 4
>>> print(variable)
0.75
>>> variable = 2 ** 3
>>> print(variable)
8
>>> variable = 5 % 2
>>> print(variable)
1
>>> variable = 5 // 2
>>> print(variable)
2
>>> variable = 5 / 2
>>> print(variable)
2.5
>>>
```

● 문자열 데이터의 표현 (이스케이프코드)

```
Python 3.11 (64-bit)
Python 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> variable = '''
... Chris
... "Hello World"
... '''
>>> print(variable)

Chris
"Hello World"

>>> variable = """Chris
... 'Hello World'"""
>>> print(variable)
Chris
'Hello World'
>>> variable = """Chris\n\"Hello World\"""
>>> print(variable)
Chris
"Hello World"
>>> 
```

● 문자열 데이터를 위한 기본 연산자

```
Python 3.11 (64-bit)
Python 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> variable = "Hello" + " " + "World"
>>> print(variable)
Hello World
>>> variable = "Hello" * 2
>>> print(variable)
HelloHello
>>> variable = len("Hello World")
>>> print(variable)
11
>>> print(variable[3])
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: 'int' object is not subscriptable
>>> variable = "Hello World"
>>> print(variable[3])
l
>>> print(variable[-3])
r
>>> print(variable[1:4])
ell
>>> print(variable[:5])
Hello
>>> print(variable[6:])
World
>>>
```

- 문자열 데이터 포맷팅 (문자열 만들기)

```
Python 3.11 (64-bit)
Python 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> variable = "%d books" % 3
>>> print(variable)
3 books
>>> variable = "%s books" % "3"
>>> print(variable)
3 books
>>> variable = "%s books" % 3
>>> print(variable)
3 books
>>> variable = "I have {0} {1}".format(3, "books")
>>> print(variable)
I have 3 books
>>> number = 3
>>> variable = f'I have {number+2} books'
>>> print(variable)
I have 5 books
>>> variable = "I have %d apples and %d books" % (number, number+1)
>>> print(variable)
I have 3 apples and 4 books
>>> variable = "The error rates : %d%%" % 55
>>> print(variable)
The error rates : 55%
>>>
```

- 문자열 데이터 포맷팅 (정렬 등 다양한 표현 방식)

Python 3.11 (64-bit)

```
Python 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> variable = "%10s" % "Hello"
>>> print(variable)
     Hello
>>> variable = "%-10s" % "Hello"
>>> print(variable)
Hello
>>> variable = "{0:<10}".format("Hello")
>>> print(variable)
Hello
>>> variable = "{0:>10}".format("Hello")
>>> print(variable)
     Hello
>>> variable = "{0:^10}".format("Hello")
>>> print(variable)
  Hello
>>> variable = "{0:-^10}".format("Hello")
>>> print(variable)
--Hello---
>>> variable = "%f" % 3.141592
>>> print(variable)
3.141592
>>> variable = "%.4f" % 3.141592
>>> print(variable)
3.1416
>>> variable = "%10.4f" % 3.141592
>>> print(variable)
    3.1416
>>>
```

2. 파이썬 입출력 함수의 활용

<*Visual Studio Code*를 이용한 화면 입출력 함수의 활용>

- inout.py

inout.py > ...

```python
# 사용자가 입력한 값은 문자열입니다.
variable = input()
print("Input String %s, Ineger %d" % (variable, int(variable)))

#사용자에게 메시지를 표시하면서 입력받을 수 있습니다.

variable = input("Enter a number: ")
variable = int(variable) ** 2
print(f'Result : {variable}')

# print를 이용한 다양한 출력 방법
variable = [1, 2, 3, 4, 5]
print(f'List : {variable}')

variable = (1, 2, 3, 4, 5)
print(f'Tuple : {variable}')

print("Hello" "World")
print("Hello", "World")

for variable in range(1, 10):
    print(variable, end=" ")
```

● 실행 결과

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL GITLENS COMMENTS Python Debug Console

```
Microsoft Windows [Version 10.0.22621.819]
(c) Microsoft Corporation. All rights reserved.

C:\_Lecture\SJCU\sjcu-python> cmd /C ""C:\Program Files\Python311\python.exe" c:\Users\chris\.vscode\extensions\ms-python.python-2023.2.0\pythonFiles\lib\python\debugpy\ada
pter/../..\debugpy\launcher 30538 -- C:\_Lecture\SJCU\sjcu-python\inout.py "
10
Input String 10, Ineger 10
Enter a number: 5
Result : 25
List : [1, 2, 3, 4, 5]
Tuple : (1, 2, 3, 4, 5)
HelloWorld
Hello World
1 2 3 4 5 6 7 8 9
C:\_Lecture\SJCU\sjcu-python>
```

- 파일 입출력 예제 : **fileinout.py**

fileinout.py > ...

```python
# 새로운 파일에 데이터 쓰기
file = open("test.txt", "w")
for data in range(1, 11):
    file.write(f'{data} line');
file.close()

# 기록한 데이터 읽기
file = open("test.txt", "r")
while True:
    line = file.readline()
    if not line:
        break
    print(line)
file.close

# 이미 존재하는 파일에 데이터 추가하기
file = open("test.txt", "a")
for data in range(1, 11):
    file.write(f'{data} line\n');
file.close()

# 기록한 데이터 읽기 (전체 라인을 한번에 읽기)
file = open("test.txt", "r")
lines = file.readlines()
for line in lines:
    print(line)
file.close()

# 기록한 데이터 읽기 (read함수를 이용하여 전체를 한번에 읽음)
file = open("test.txt", "r")
data = file.read()
print(data)
file.close()

# with ~ as 구문을 이용하여 간결한 코드 작성하기
with open("test.txt", "r") as file:
    data = file.read()
    print(data)
```

● 실행 결과

```
1 line2 line3 line4 line5 line6 line7 line8 line9 line10 line
1 line2 line3 line4 line5 line6 line7 line8 line9 line10 line1 line

2 line

3 line

4 line

5 line

6 line

7 line
8 line
9 line
10 line

1 line2 line3 line4 line5 line6 line7 line8 line9 line10 line1 line
2 line
3 line
4 line
5 line
6 line
7 line
8 line
9 line
10 line
```

※ **CMD (**명령창**)**에서 파이썬 코드 실행하기

```
C:\_Lecture\SJCU\sjcu-python>python fileinout.py
1 line2 line3 line4 line5 line6 line7 line8 line9 line10 line
1 line2 line3 line4 line5 line6 line7 line8 line9 line10 line1 line

2 line

3 line

4 line

5 line

6 line

7 line

8 line

9 line

10 line

1 line2 line3 line4 line5 line6 line7 line8 line9 line10 line1 line
2 line
3 line
4 line
5 line
6 line
7 line
8 line
9 line
10 line

1 line2 line3 line4 line5 line6 line7 line8 line9 line10 line1 line
2 line
3 line
4 line
5 line
6 line
7 line
8 line
9 line
10 line
```

“끝”

감사합니다.